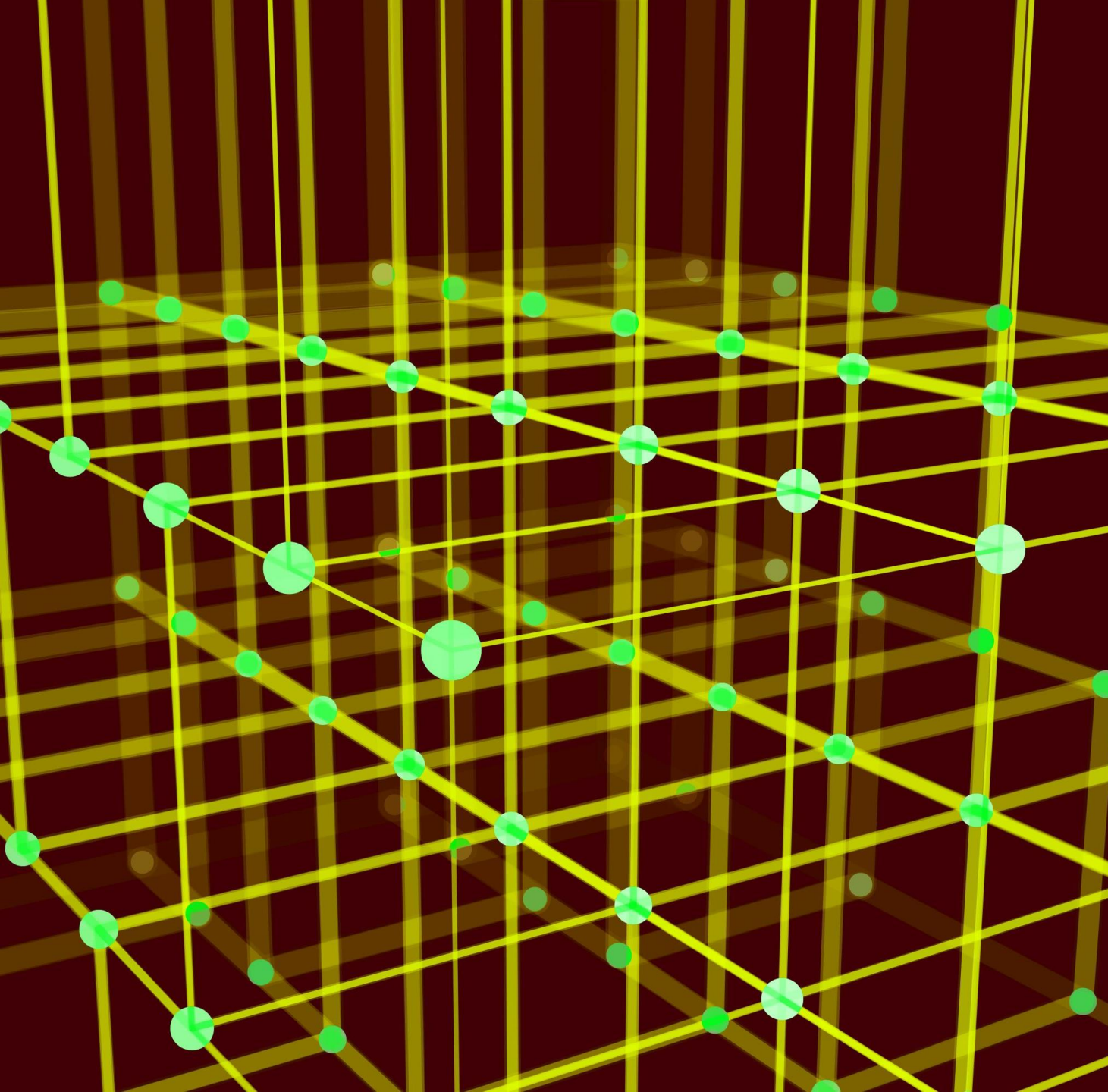




Hashlite – FPGA Cryptography

Gavin Walsh – G00402296

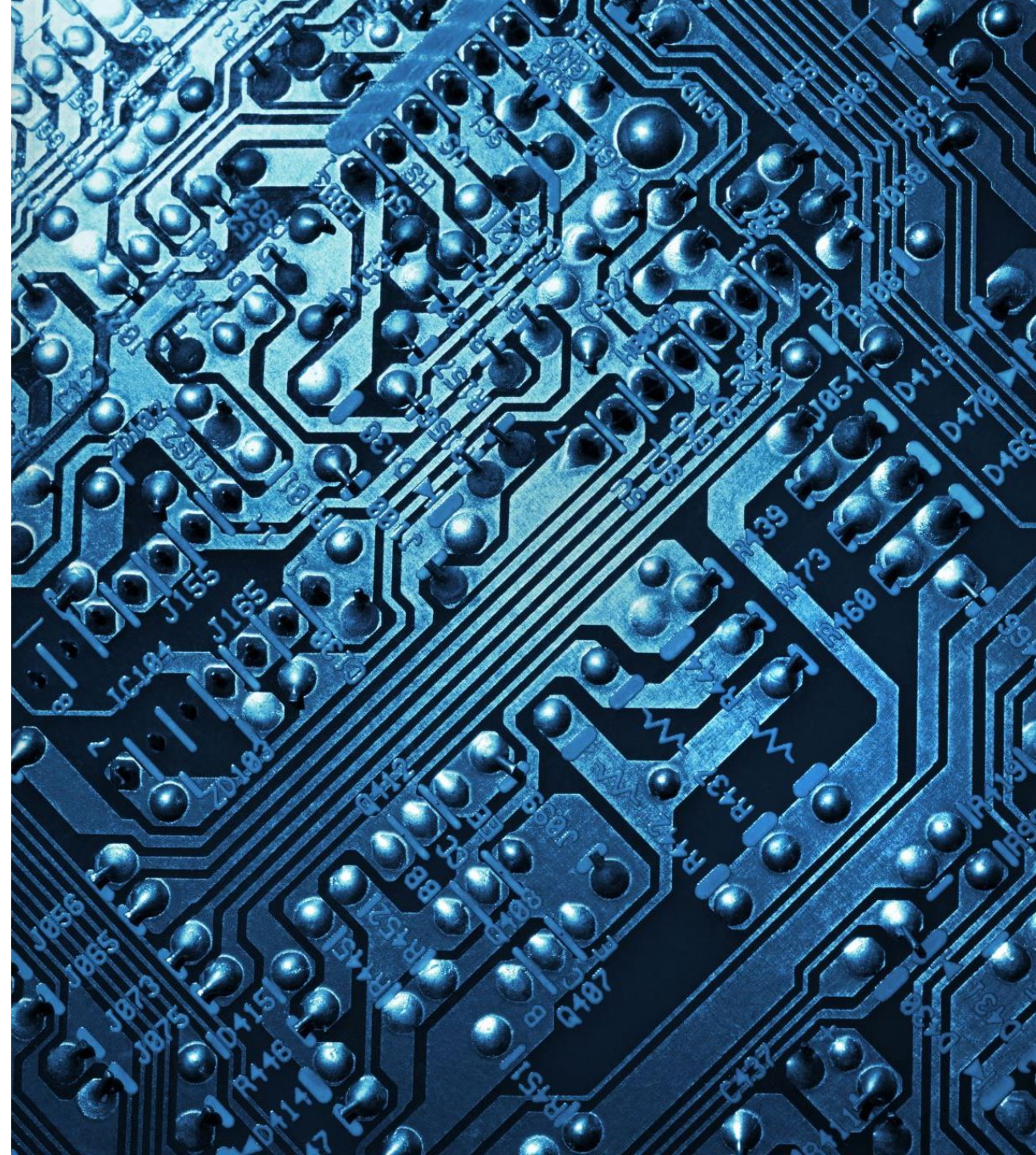
Introduction

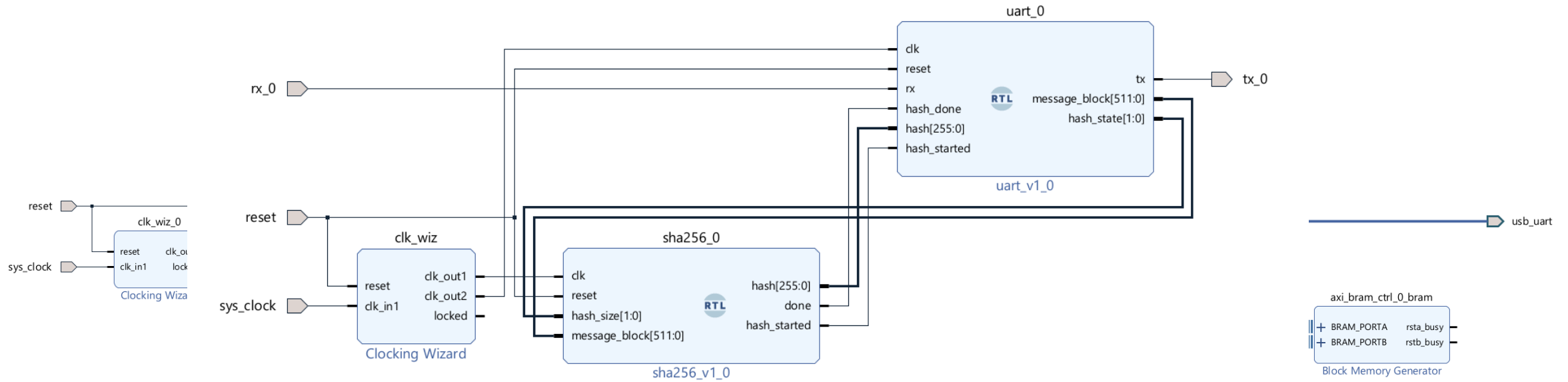
What?

Implemented a hardware-accelerated SHA-256 core on a Xilinx Artix-7 FPGA.

Why?

To demonstrate the performance advantages of a hardware implementation over traditional software for computationally intensive task.

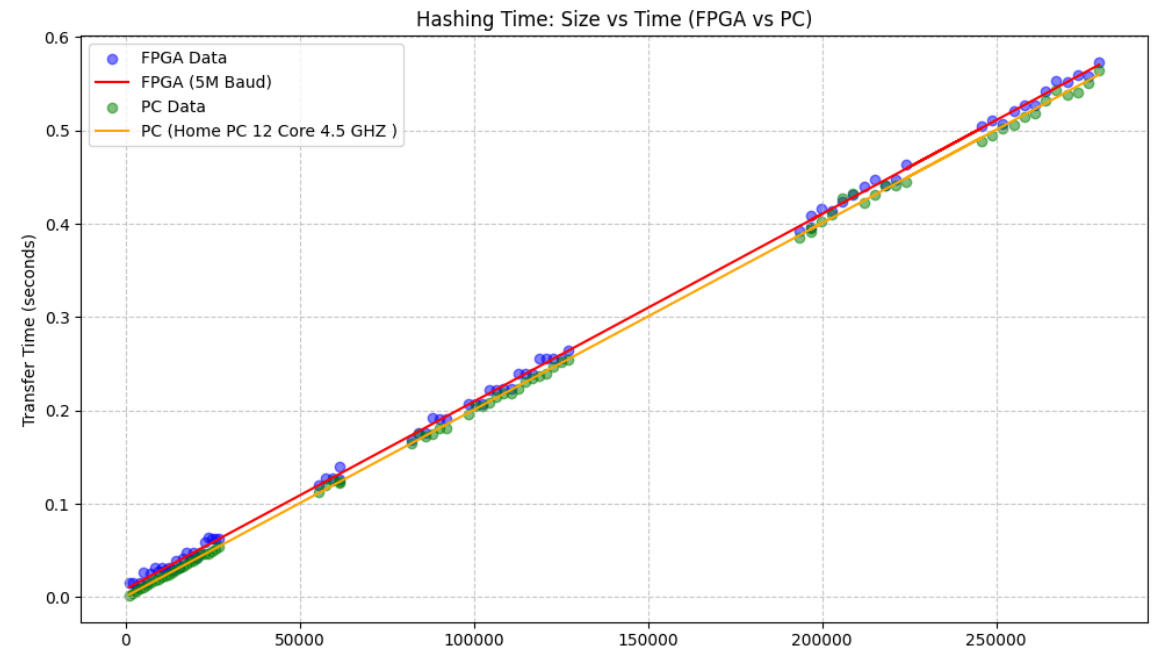
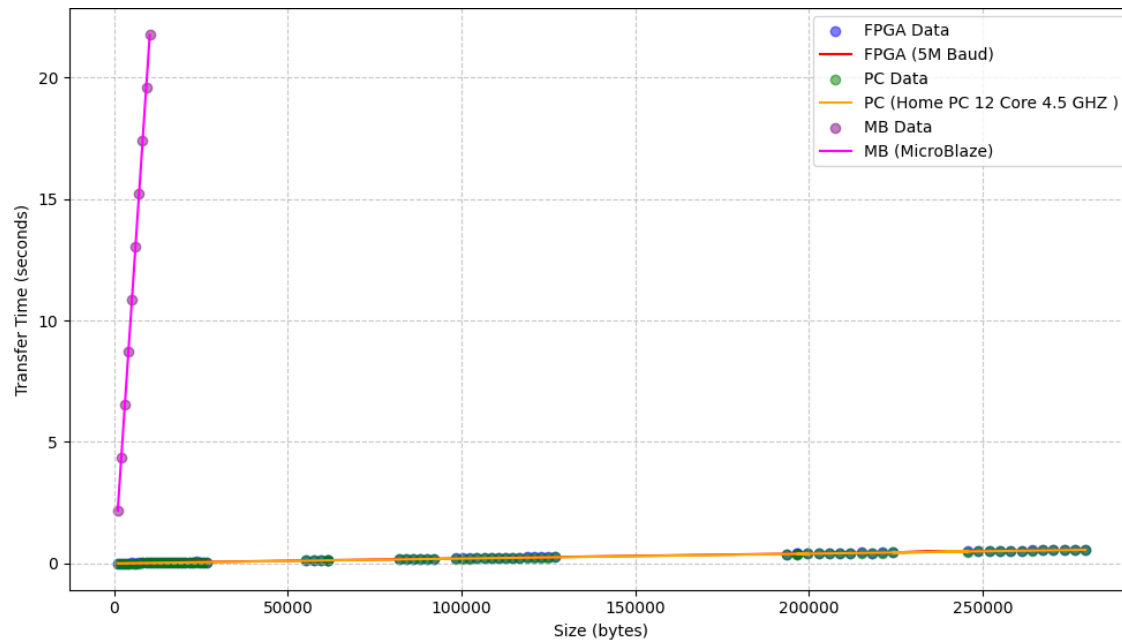




System Architecture

Key Components

- Sha256 Module – The Accelerator Core
 - Clocked at 50mhz. Capable of 770,000 hashes a second
- UART Module – For Data Input and Output
 - Running at 5,000,000 baud.



Results

- New Implementation 160 hash's a second.
- Old Implematation 0.45 hash's a second.

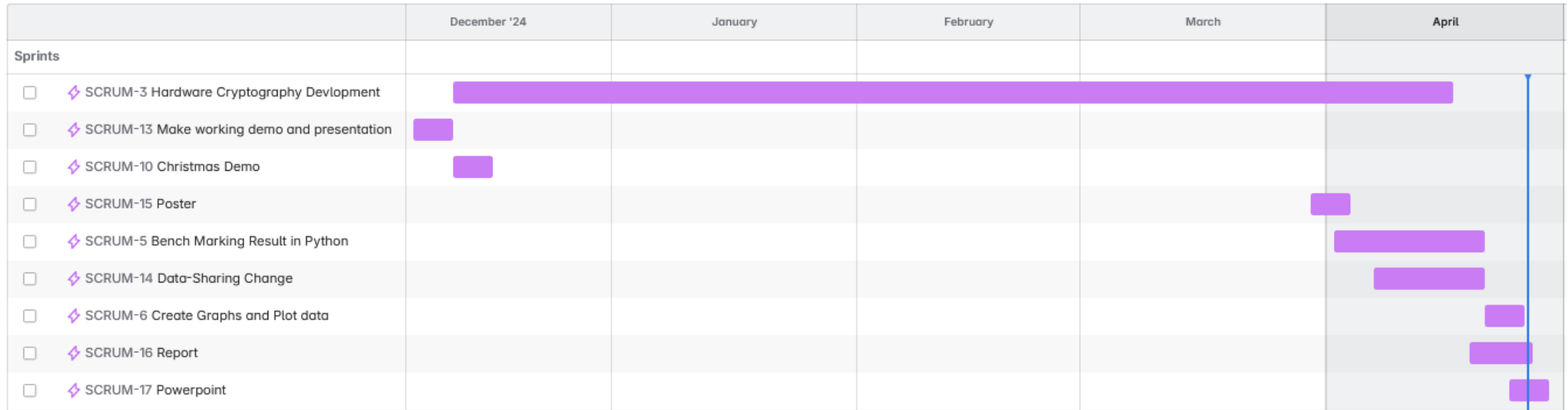
Technologies

Languages Used:

- Verilog
- Python

Tools Used:

- Vivado
- PyCharm
- Jira
- OneNote



Project Management

Project Management Tool: Jira

Scrum Philosophy: Kanban



Demo Time!
Thank you for listening.